Meta-analysis technical report

Evolution of ISUP Grade Group 1 Detection Across Prostate Cancer Screening Strategies

Analysis: meta_analysis.R · Numeric log: output/results.txt · Source: primary-outcome table

R 4.6.0; `meta` 8.5.0; `metafor` 5.0.1.

1. Data and analysis units

The extraction contains nine randomized trials and 18 arms. Comparative models use paired randomized arms; proportion models use 16 study-strategy units.

Cohort	Included trials	Use
Screening invitation vs no organised screening	ERSPC, PLCO, CAP, ProScreen	Pooled comparative RRs
Diagnostic/biopsy strategy; both arms screened	STHLM3-MRI, MVP, GÖTEBORG-2, PROSA	Pooled comparative RRs
Timing of screening initiation	PROBASE	Standalone estimate

GÖTEBORG-2 arms assigned to the same broad `PSA → MRI` category are combined. Different strategies within a trial remain separate. PROBASE contributes only its immediate PSA arm to strategy proportions. Its delayed comparator included 6,537 DRE assessments and two cancers, but no coherent PSA-screened denominator at this endpoint.

CAP randomized 573 practices. No ICC or cluster-adjusted grade-specific variance was available, so its confidence intervals and weights are unadjusted; all invitation-family outcomes are also analyzed without CAP.

Grade-yield inputs

`NR` denotes no coherent reported denominator.

Trial	GG1 / biopsied, I	GG1 / biopsied, C	GG≥2 / biopsied, I	GG≥2 / biopsied, C
ERSPC	4,325 / 17,543 (24.7%)	NR	1,665 / 17,543 (9.5%)	NR
PLCO	NR	NR	NR	NR
CAP	NR	NR	NR	NR
ProScreen	39 / 262 (14.9%)	NR	172 / 262 (65.6%)	NR
PROBASE	15 / 120 (12.5%)	2 / 37 (5.4%)	33 / 120 (27.5%)	0 / 37 (0.0%)
STHLM3-MRI	34 / 311 (10.9%)	73 / 438 (16.7%)	192 / 331 (58.0%)	106 / 438 (24.2%)
MVP	4 / 24 (16.7%)	4 / 28 (14.3%)	11 / 24 (45.8%)	4 / 28 (14.3%)
GÖTEBORG-2	68 / 254 (26.8%)	159 / 588 (27.0%)	117 / 254 (46.1%)	139 / 588 (23.6%)
PROSA	9 / 40 (22.5%)	4 / 20 (20.0%)	17 / 40 (42.5%)	7 / 20 (35.0%)

Trial	GG≥2 / all detected cancers, I	GG≥2 / all detected cancers, C
ERSPC	1,665 / 5,990 (27.8%)	1,947 / 4,307 (45.2%)
PLCO	1,104 / 3,452 (32.0%)	1,120 / 2,974 (37.7%)
CAP	6,649 / 12,013 (55.3%)	8,019 / 12,958 (61.9%)
ProScreen	172 / 219 (78.5%)	282 / 355 (79.4%)
PROBASE	33 / 48 (68.8%)	0 / 2 (0.0%)
STHLM3-MRI	192 / 226 (85.0%)	106 / 179 (59.2%)
MVP	11 / 15 (73.3%)	4 / 8 (50.0%)
GÖTEBORG-2	117 / 185 (63.2%)	139 / 298 (46.6%)
PROSA	17 / 26 (65.4%)	7 / 11 (63.6%)

CAP is excluded from grade-per-biopsy analyses because its 5,848 biopsies concern the first screening round, while grade counts are cumulative over 15 years. STHLM3-MRI reports intervention GG≥2 as `192/331`, although its general biopsy count is 311; the outcome-specific denominator is retained.

2. Analysis specification

- Detection-rate RRs and proportions use randomized/invited denominators.
- Cancer-grade composition uses all detected prostate cancers.
- Biopsy yield uses outcome-specific biopsy denominators.
- Biopsy use is expressed per screened/assessed participant.
- Invitation and diagnostic-strategy cohorts are fitted separately.
- Strategy proportions are fitted separately and are not head-to-head comparisons.
- A comparative biopsy-yield RR requires usable denominators in both arms.

Models:

- Risk ratios: inverse-variance random effects, DerSimonian–Laird variance, Hartung–Knapp confidence intervals, and a 0.5 all-cell correction for zero cells.
- Proportions: random-effects logit-normal GLMMs; exact binomial intervals for single-study groups.
- Prediction intervals are reported for models with at least three studies.
- Leave-one-out analysis omits each eligible contribution once within its cohort or strategy.

3. Comparative risk ratios

3.1 GG1 detection per randomized/invited participant

Cohort	k	RR (95% CI)	I²	95% prediction interval
Invitation vs usual care	4	1.614 (1.049–2.483)	99.0%	0.533–4.887
Diagnostic/biopsy strategy	4	0.563 (0.152–2.081)	74.6%	0.093–3.397
PROBASE timing contrast	1	7.487 (1.712–32.736)	—	—

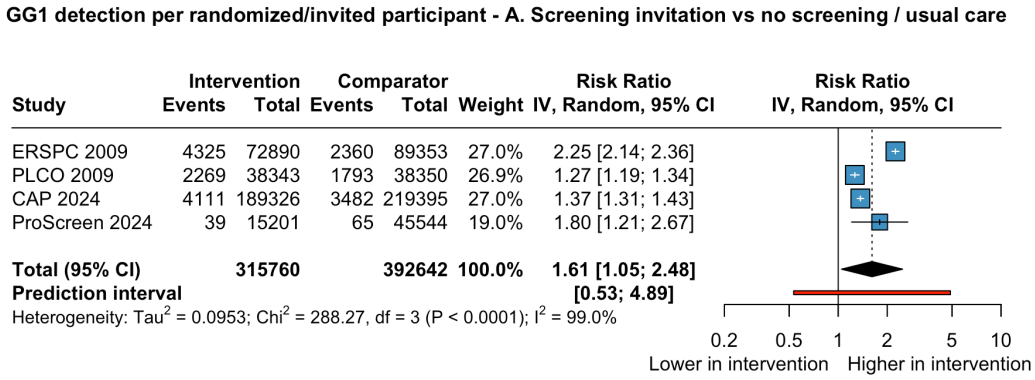


Figure 1a. Forest plot for GG1 detection per randomized/invited participant — screening invitation vs no organised screening.

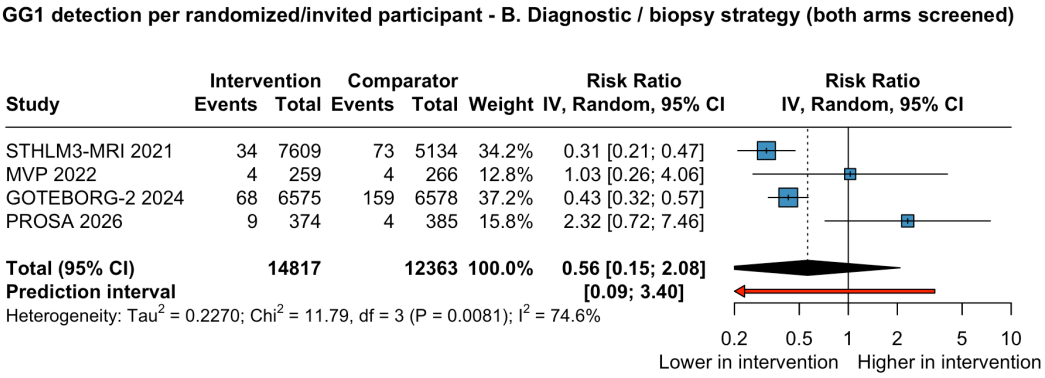


Figure 1b. Forest plot for GG1 detection per randomized/invited participant — diagnostic/biopsy strategy.

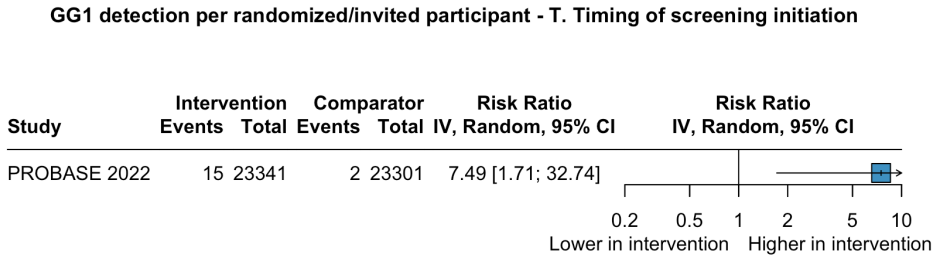


Figure 1c. Forest plot for GG1 detection per randomized/invited participant — PROBASE timing-of-screening contrast.

3.2 GG1 among all detected prostate cancers

Cohort	k	RR (95% CI)	I²	95% prediction interval
Invitation vs usual care	4	1.204 (1.003–1.444)	95.2%	0.840–1.725
Diagnostic/biopsy strategy	4	0.567 (0.302–1.064)	68.9%	0.152–2.113
PROBASE timing contrast	1	0.380 (0.198–0.729)	—	—

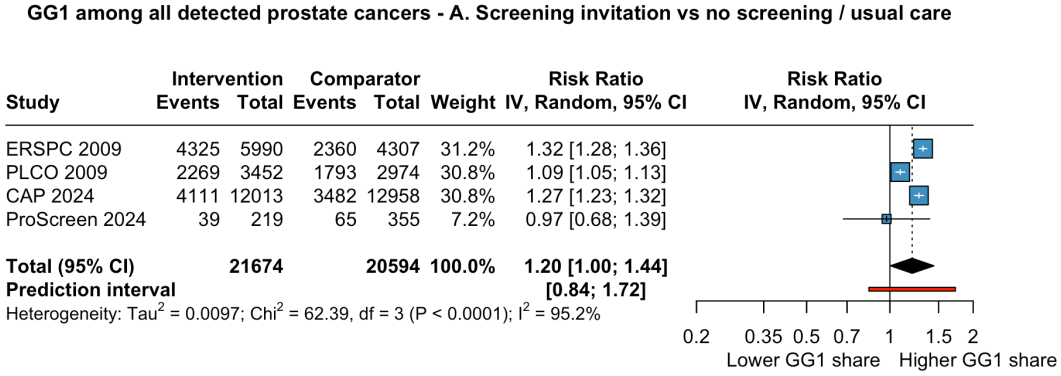


Figure 2a. Forest plot for GG1 among all detected prostate cancers — screening invitation vs no organised screening.

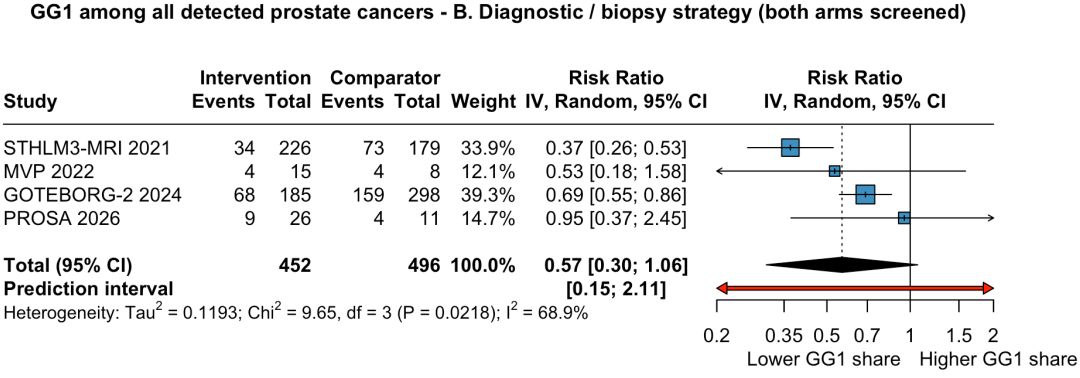


Figure 2b. Forest plot for GG1 among all detected prostate cancers — diagnostic/biopsy strategy.

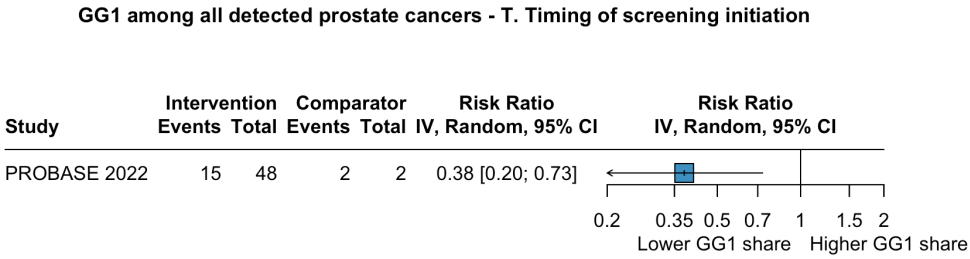


Figure 2c. Forest plot for GG1 among all detected prostate cancers — PROBASE timing-of-screening contrast.

3.3 GG≥2 detection per randomized/invited participant

Cohort	k	RR (95% CI)	I²	95% prediction interval
Invitation vs usual care	4	1.117 (0.721–1.732)	93.6%	0.692–1.802
Diagnostic/biopsy strategy	4	1.292 (0.571–2.925)	73.8%	0.373–4.475
PROBASE timing contrast	1	66.885 (4.098–1091.529)	—	—

The PROBASE comparator has zero inferred GG≥2 events; continuity-correction sensitivity is reported below.

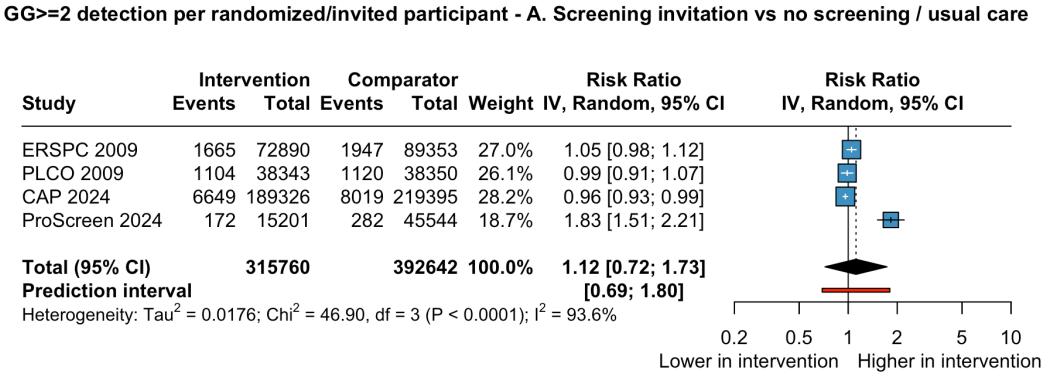


Figure 3a. Forest plot for GG≥2 detection per randomized/invited participant — screening invitation vs no organised screening.

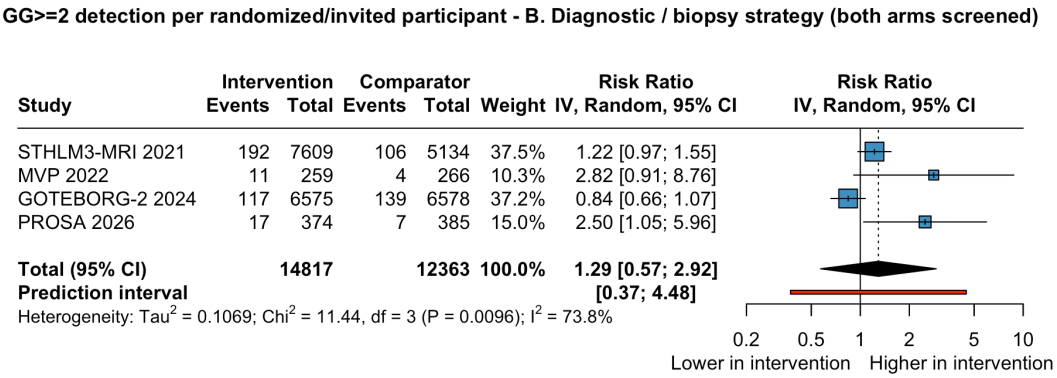


Figure 3b. Forest plot for GG≥2 detection per randomized/invited participant — diagnostic/biopsy strategy.

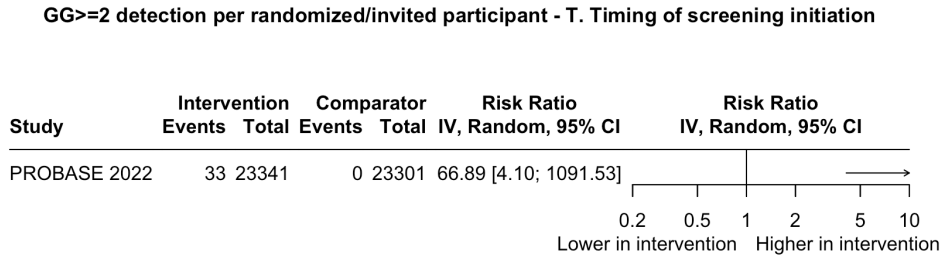


Figure 3c. Forest plot for GG≥2 detection per randomized/invited participant — PROBASE timing-of-screening contrast.

3.4 GG≥2 among all detected prostate cancers

Trial	Intervention strategy: GG≥2 / all cancers	Comparator strategy: GG≥2 / all cancers	Trial RR
STHLM3-MRI	Biomarker → MRI: 192/226 (85.0%)	PSA: 106/179 (59.2%)	1.435
MVP	MRI-first: 11/15 (73.3%)	PSA: 4/8 (50.0%)	1.467
GÖTEBORG-2	MRI-targeted only: 117/185 (63.2%)	Systematic ± targeted: 139/298 (46.6%)	1.356
PROSA	MRI-first: 17/26 (65.4%)	PSA-triggered MRI: 7/11 (63.6%)	1.027

For STHLM3-MRI, 85.0% / 59.2% gives RR 1.435. The pooled RR is 1.388, equivalent to a 38.8% higher GG≥2 share among detected cancers.

Cohort	k	RR (95% CI)	I²	95% prediction interval
Invitation vs usual care	4	0.823 (0.594–1.141)	98.3%	0.409–1.659
Diagnostic/biopsy strategy	4	1.388 (1.233–1.563)	0.0%	1.179–1.634
PROBASE timing contrast	1	4.102 (0.324–51.881)	—	—

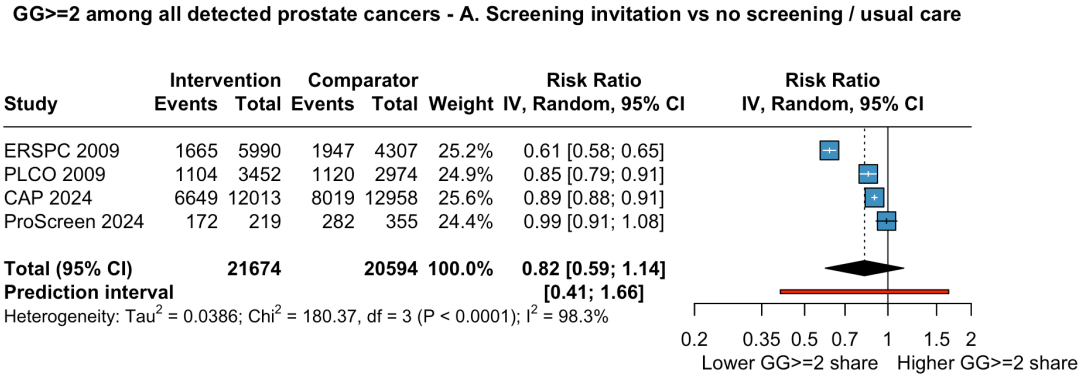


Figure 4a. Forest plot for GG≥2 among all detected prostate cancers — screening invitation vs no organised screening.

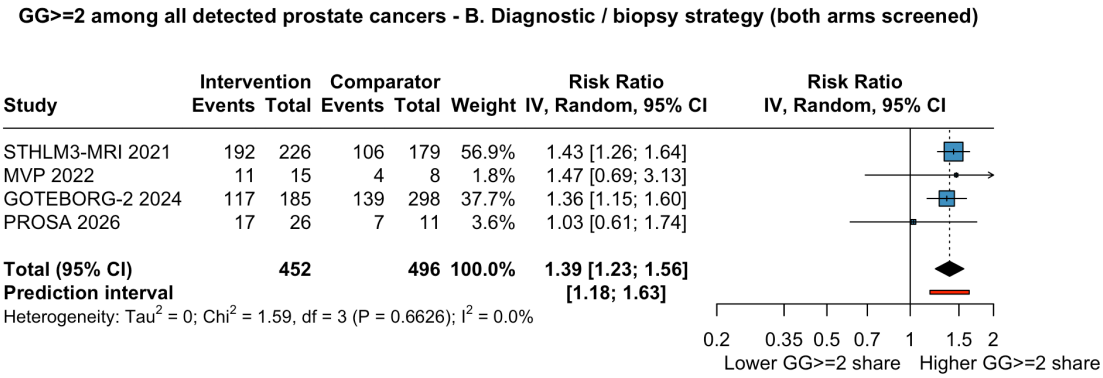


Figure 4b. Forest plot for GG≥2 among all detected prostate cancers — diagnostic/biopsy strategy.

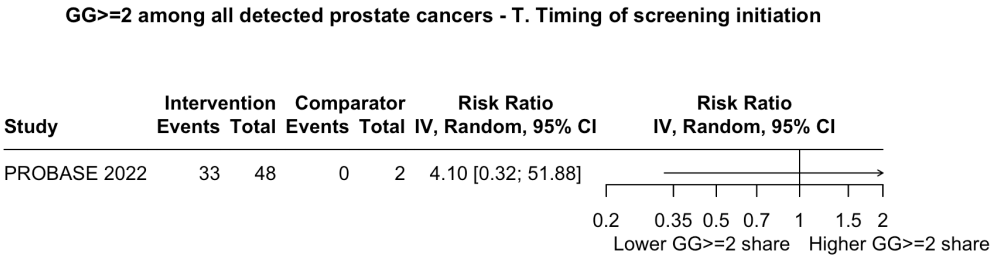


Figure 4c. Forest plot for GG≥2 among all detected prostate cancers — PROBASE timing-of-screening contrast.

3.5 GG1 among participants biopsied

Cohort	k	RR (95% CI)	I²	95% prediction interval
Diagnostic/biopsy strategy	4	0.878 (0.601–1.281)	16.2%	0.512–1.504
PROBASE timing contrast	1	2.313 (0.554–9.650)	—	—

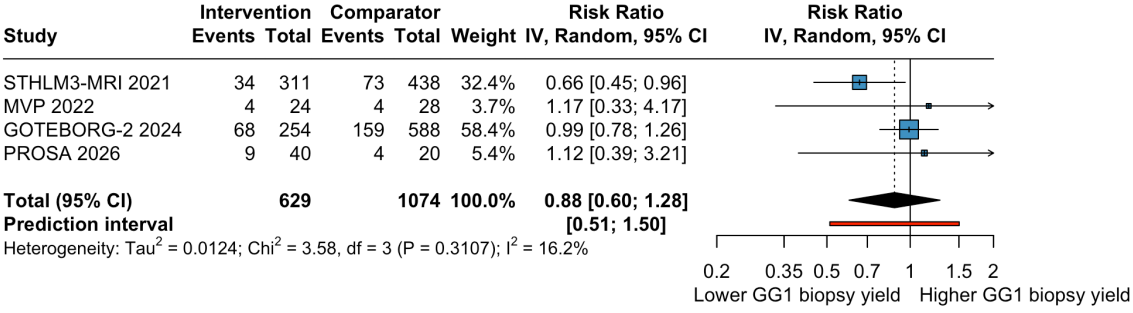


Figure 5a. Forest plot for GG1 among participants biopsied — diagnostic/biopsy strategy.

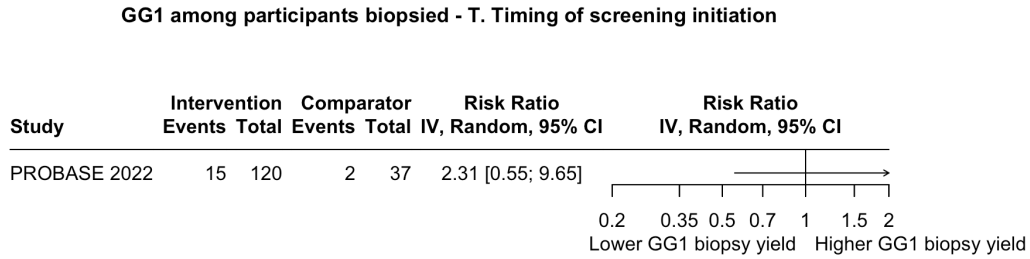


Figure 5b. Forest plot for GG1 among participants biopsied — PROBASE timing-of-screening contrast.

3.6 GG≥2 among participants biopsied

Cohort	k	RR (95% CI)	I^2	95% prediction interval
Diagnostic/biopsy strategy	4	2.094 (1.421–3.087)	44.5%	1.175–3.732
PROBASE timing contrast	1	21.041 (1.320–335.297)	—	—

No invitation-family biopsy-yield RR is estimable.

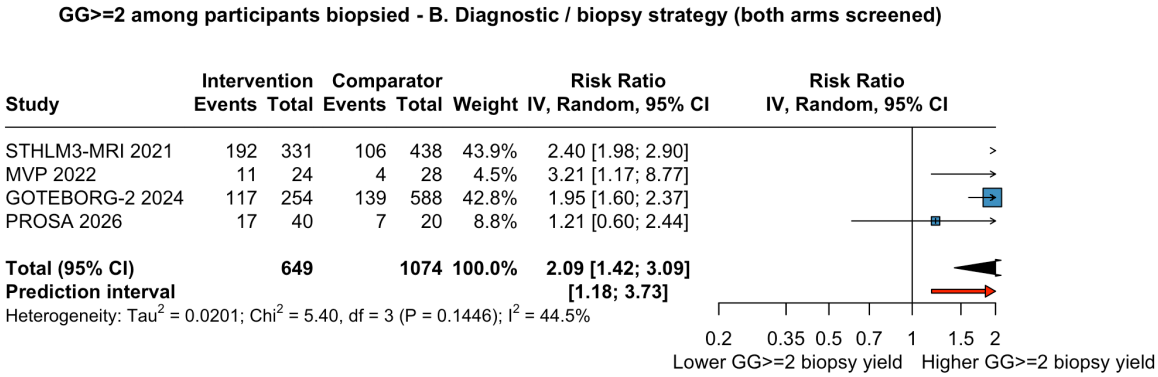


Figure 6a. Forest plot for GG≥2 among participants biopsied — diagnostic/biopsy strategy.

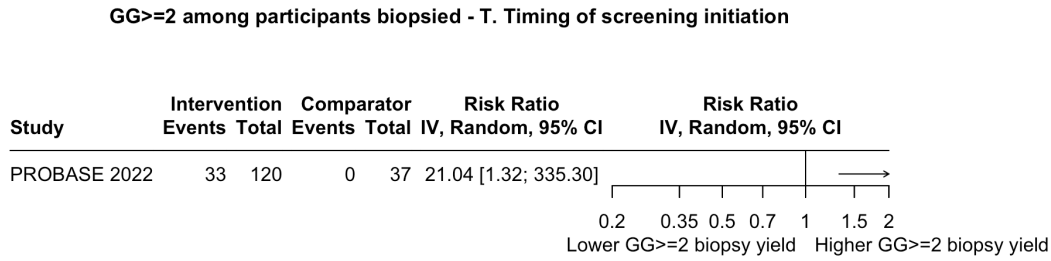


Figure 6b. Forest plot for GG≥2 among participants biopsied — PROBASE timing-of-screening contrast.

4. Proportions by screening strategy

These are separate descriptive models, not randomized head-to-head comparisons. `k` is the number of study-strategy units.

4.1 GG1 per randomized/invited participant

Strategy	k	Proportion (95% CI)	I²
No screening / usual care	4	1.309% (0.355–4.706)	99.8%
PSA + DRE	1	5.918% (5.684–6.159)	—
PSA	5	1.112% (0.287–4.217)	99.8%
PSA → MRI	2	1.706% (1.501–1.939)	2.9%
Biomarker → MRI	2	0.332% (0.226–0.487)	82.2%
MRI-first	2	2.054% (1.196–3.504)	0.0%

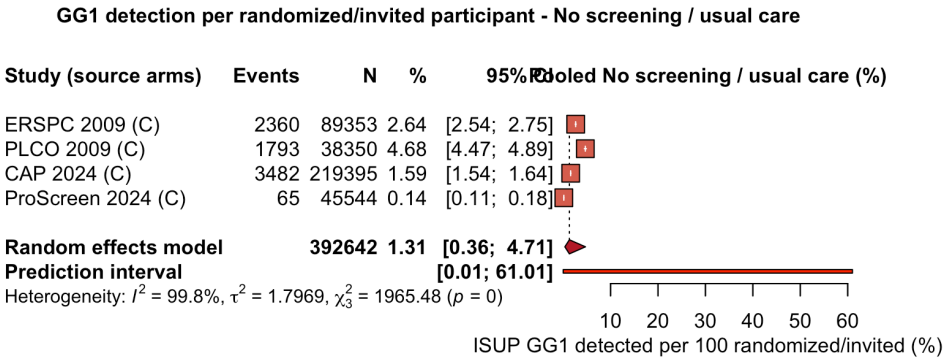


Figure 7a. Forest plot for GG1 per randomized/invited participant — no screening / usual care.

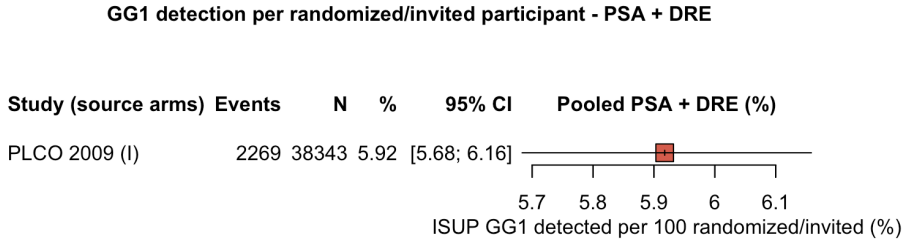


Figure 7b. Forest plot for GG1 per randomized/invited participant — PSA + DRE.

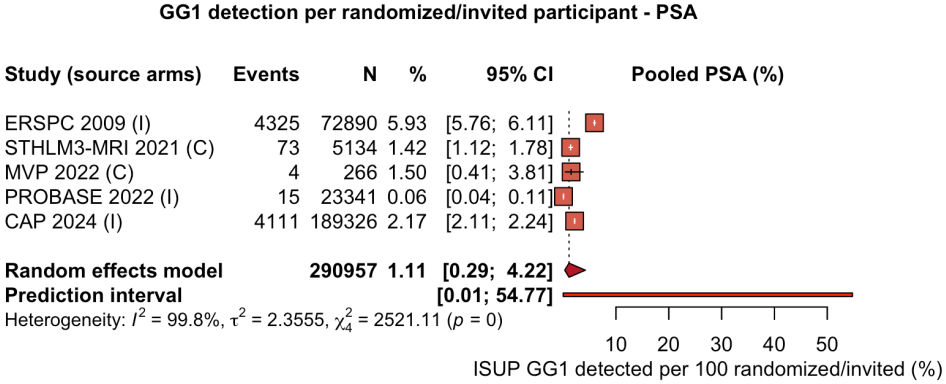


Figure 7c. Forest plot for GG1 per randomized/invited participant — PSA.

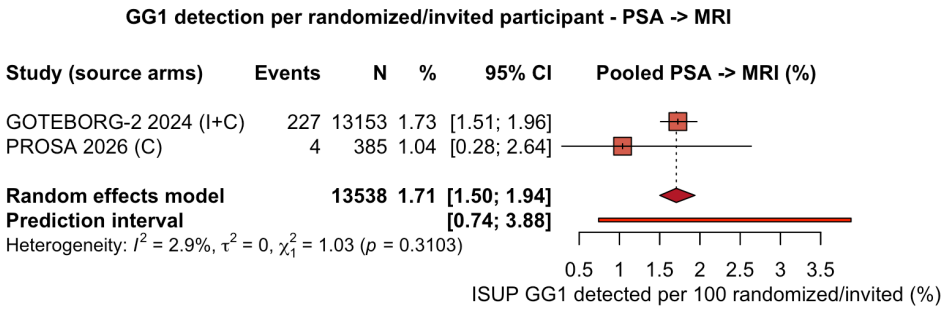


Figure 7d. Forest plot for GG1 per randomized/invited participant — PSA → MRI.

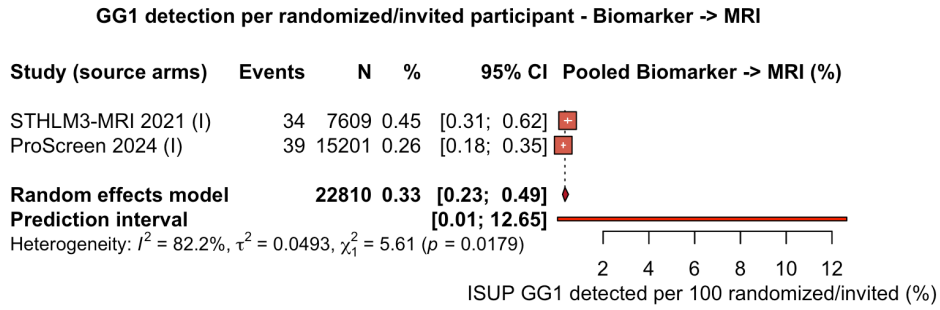


Figure 7e. Forest plot for GG1 per randomized/invited participant — biomarker → MRI.

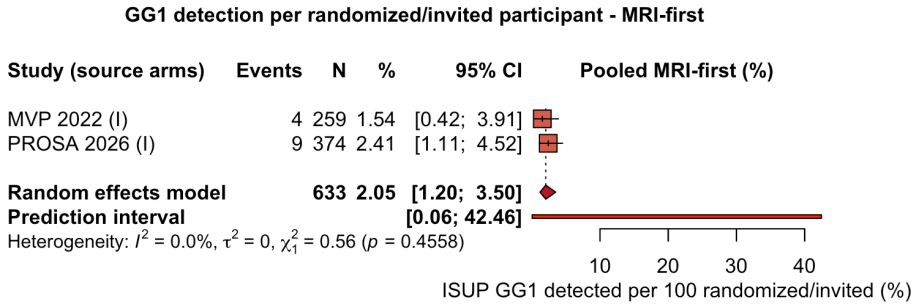


Figure 7f. Forest plot for GG1 per randomized/invited participant — MRI-first.

4.2 GG1 among all detected prostate cancers

Strategy	k	Proportion (95% CI)	I²
No screening / usual care	4	38.549% (22.295–57.834)	99.8%
PSA + DRE	1	65.730% (64.120–67.314)	—
PSA	5	45.788% (31.125–61.218)	99.8%
PSA → MRI	2	46.761% (42.398–51.175)	0.0%
Biomarker → MRI	2	16.405% (13.247–20.141)	0.0%
MRI-first	2	31.707% (19.387–47.266)	0.0%

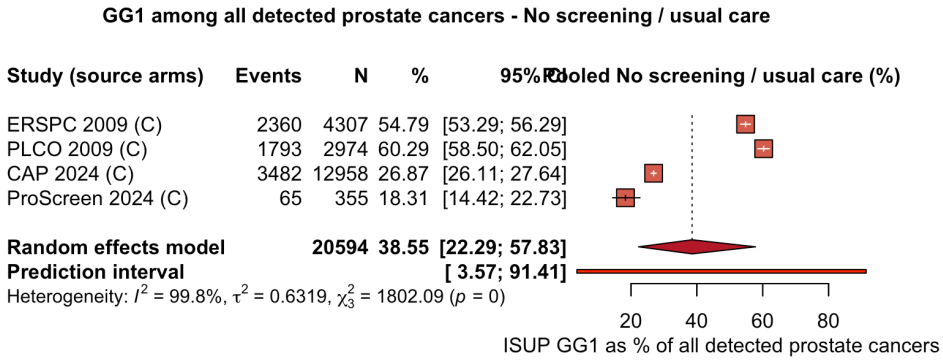


Figure 8a. Forest plot for GG1 among all detected prostate cancers — no screening / usual care.

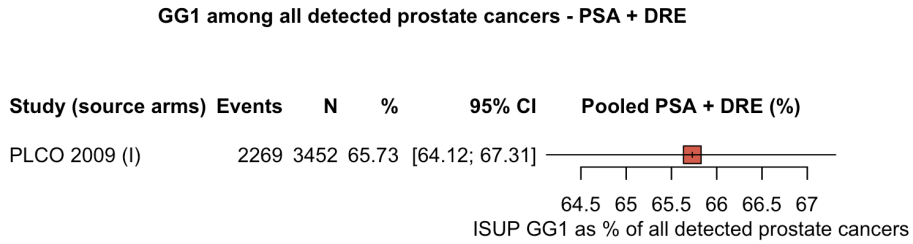


Figure 8b. Forest plot for GG1 among all detected prostate cancers — PSA + DRE.

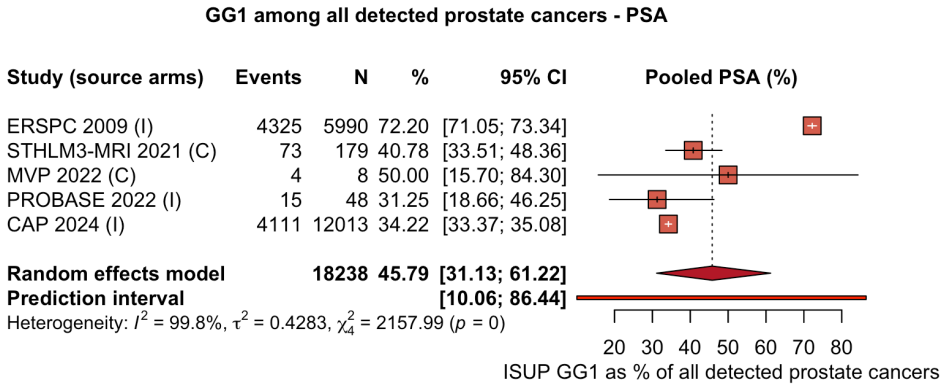


Figure 8c. Forest plot for GG1 among all detected prostate cancers — PSA.

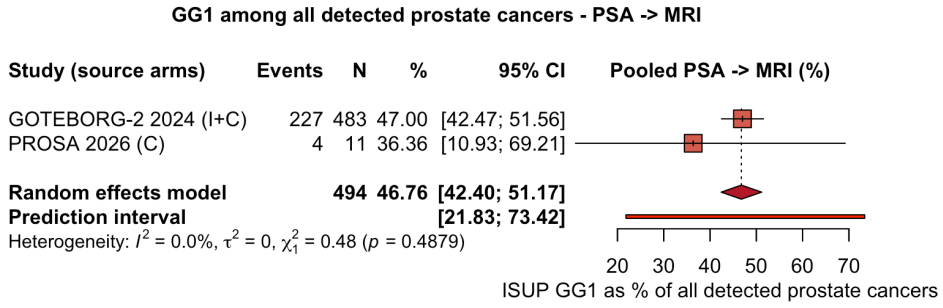


Figure 8d. Forest plot for GG1 among all detected prostate cancers — PSA → MRI.

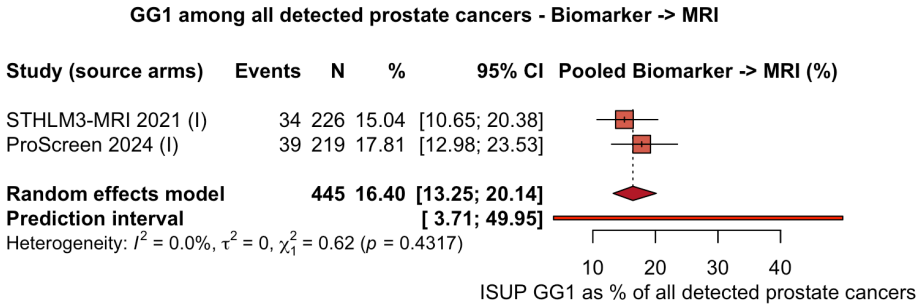


Figure 8e. Forest plot for GG1 among all detected prostate cancers — biomarker → MRI.

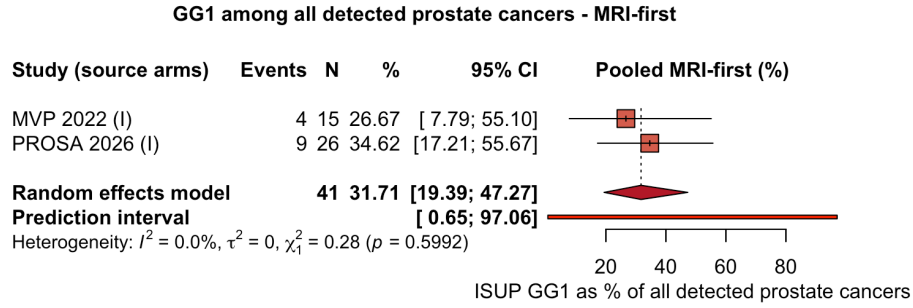


Figure 8f. Forest plot for GG1 among all detected prostate cancers — MRI-first.

4.3 GG≥2 per randomized/invited participant

Strategy	k	Proportion (95% CI)	I²
No screening / usual care	4	1.958% (0.999–3.803)	99.8%
PSA + DRE	1	2.879% (2.714–3.052)	—
PSA	5	1.271% (0.456–3.491)	99.3%
PSA → MRI	2	1.943% (1.723–2.189)	0.0%
Biomarker → MRI	2	1.689% (0.966–2.936)	98.3%
MRI-first	2	4.423% (3.071–6.332)	0.0%

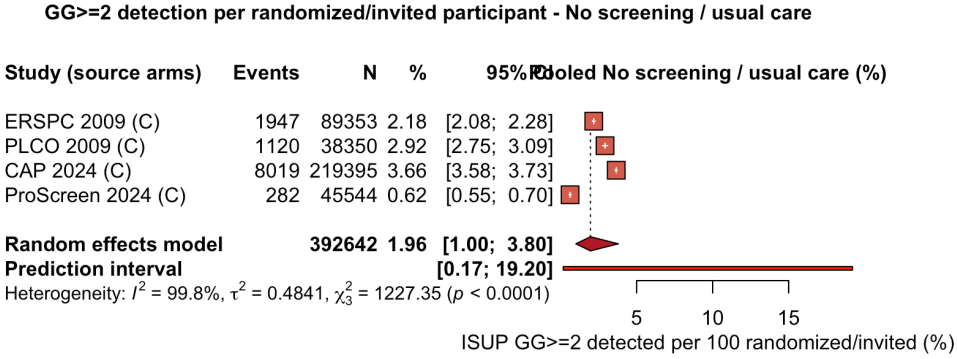


Figure 9a. Forest plot for GG≥2 per randomized/invited participant — no screening / usual care.

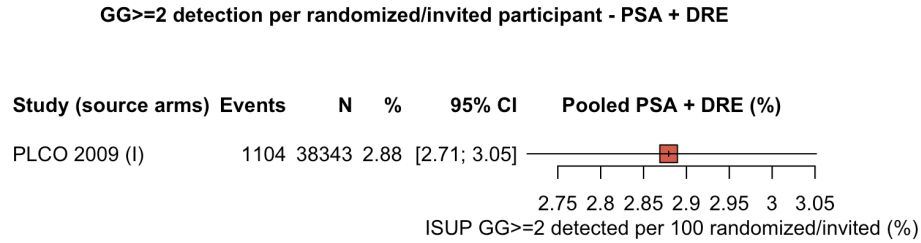


Figure 9b. Forest plot for GG≥2 per randomized/invited participant — PSA + DRE.

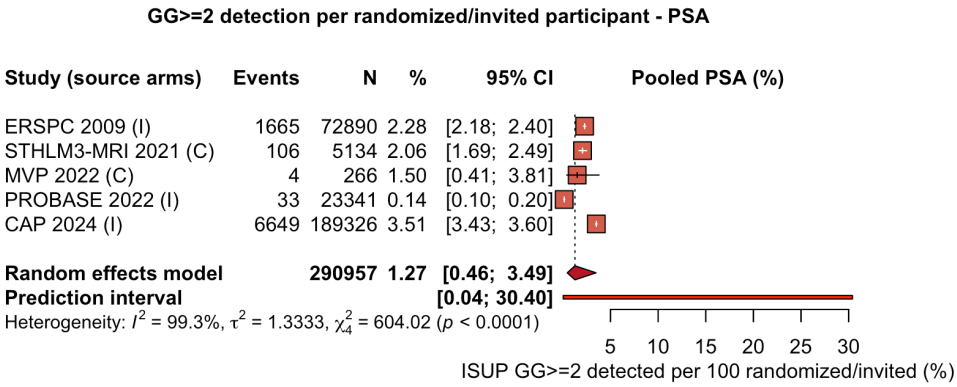


Figure 9c. Forest plot for GG≥2 per randomized/invited participant — PSA.

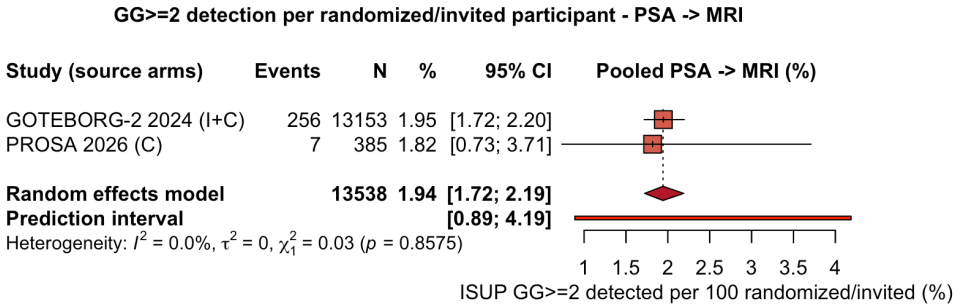


Figure 9d. Forest plot for GG≥2 per randomized/invited participant — PSA → MRI.

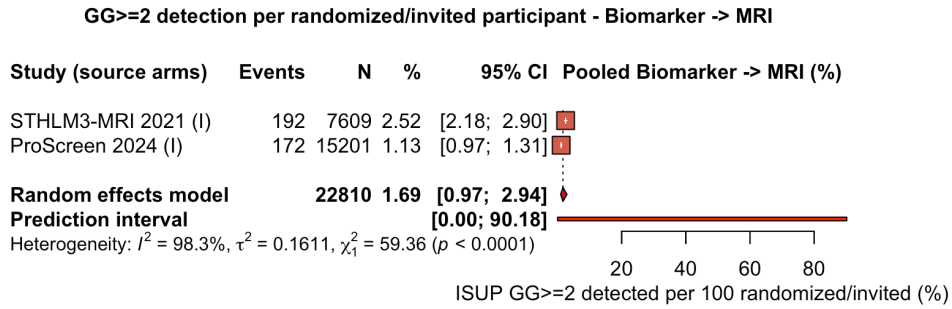


Figure 9e. Forest plot for GG≥2 per randomized/invited participant — biomarker → MRI.

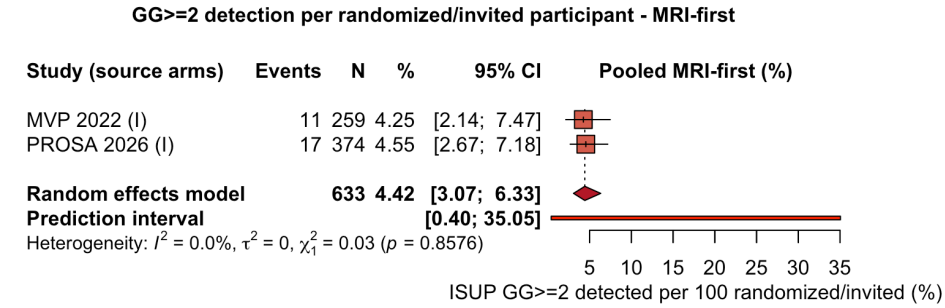


Figure 9f. Forest plot for GG≥2 per randomized/invited participant — MRI-first.

4.4 GG≥2 among all detected prostate cancers

Strategy	k	Proportion (95% CI)	I²
No screening / usual care	4	56.901% (39.783–72.515)	99.6%
PSA + DRE	1	31.982% (30.427–33.567)	—
PSA	5	51.494% (37.320–65.432)	99.7%
PSA → MRI	2	53.239% (48.825–57.602)	0.0%
Biomarker → MRI	2	81.858% (76.967–85.900)	67.2%
MRI-first	2	68.293% (52.734–80.613)	0.0%

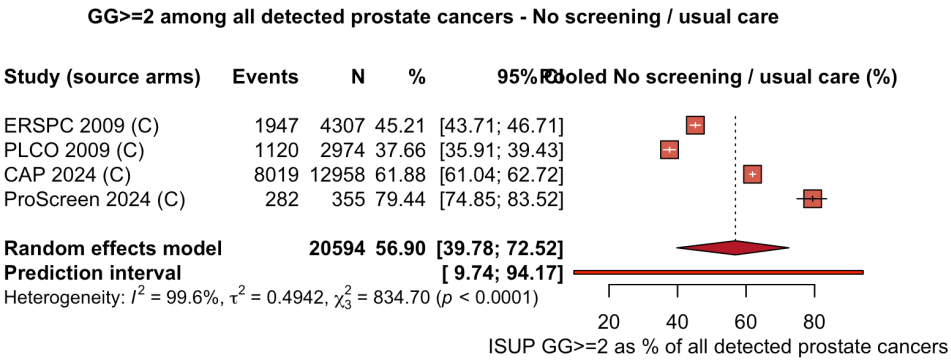


Figure 10a. Forest plot for GG≥2 among all detected prostate cancers — no screening / usual care.

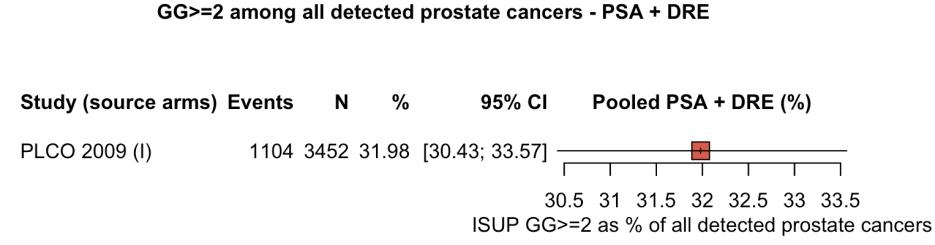


Figure 10b. Forest plot for GG≥2 among all detected prostate cancers — PSA + DRE.

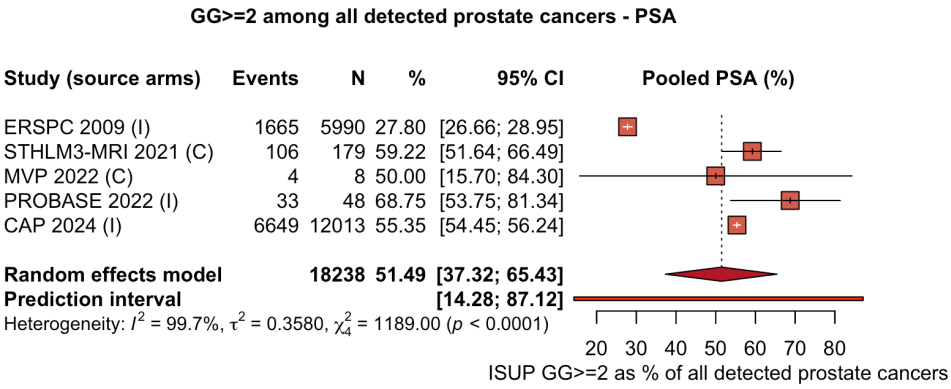


Figure 10c. Forest plot for GG≥2 among all detected prostate cancers — PSA.

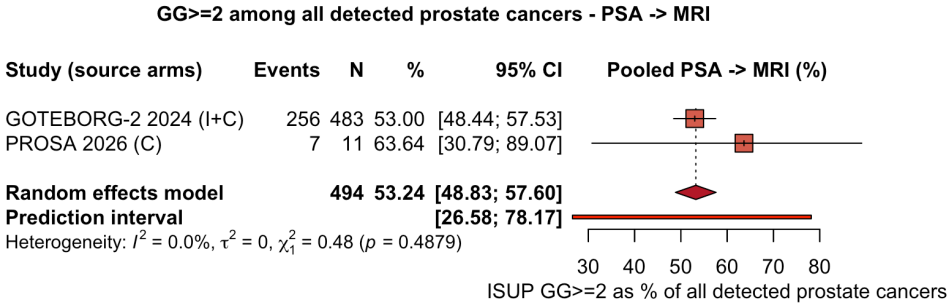


Figure 10d. Forest plot for GG≥2 among all detected prostate cancers — PSA → MRI.

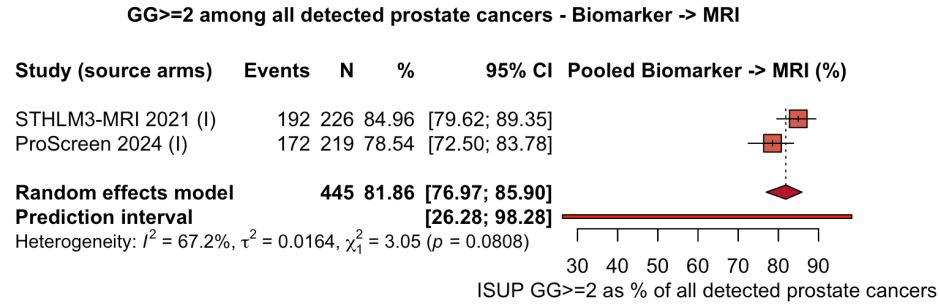


Figure 10e. Forest plot for GG≥2 among all detected prostate cancers — biomarker → MRI.

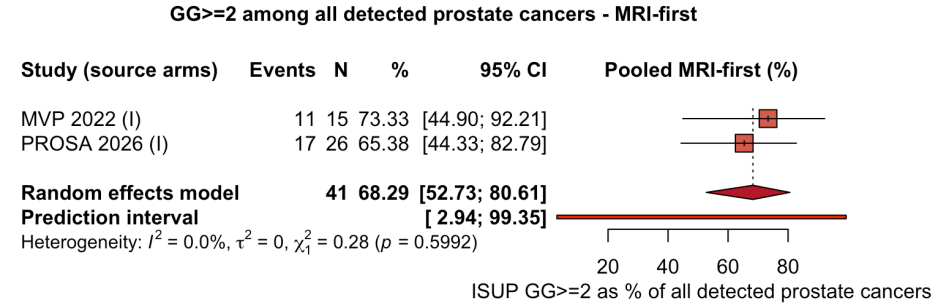


Figure 10f. Forest plot for GG≥2 among all detected prostate cancers — MRI-first.

4.5 GG1 among participants biopsied

Strategy	k	Proportion (95% CI)	I²
PSA	4	18.003% (13.108–24.217)	87.9%
PSA → MRI	2	26.798% (23.947–29.856)	0.0%
Biomarker → MRI	2	12.740% (10.250–15.728)	49.6%
MRI-first	2	20.313% (12.177–31.910)	0.0%

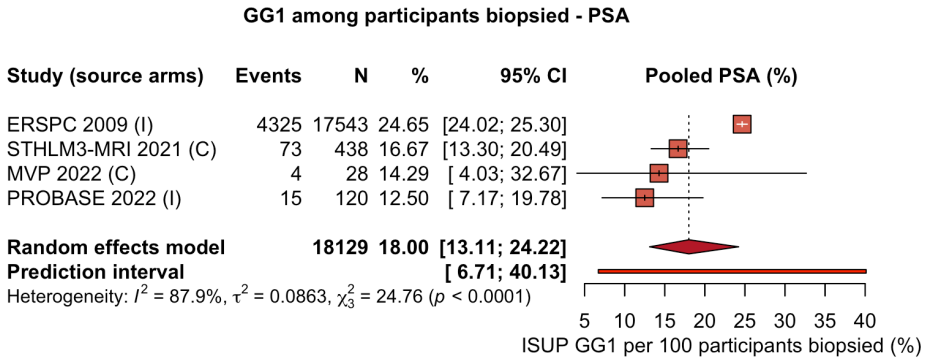


Figure 11a. Forest plot for GG1 among participants biopsied — PSA.

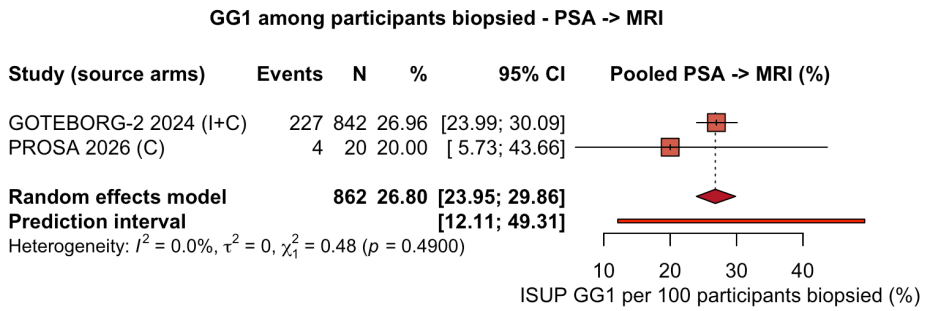


Figure 11b. Forest plot for GG1 among participants biopsied — PSA → MRI.

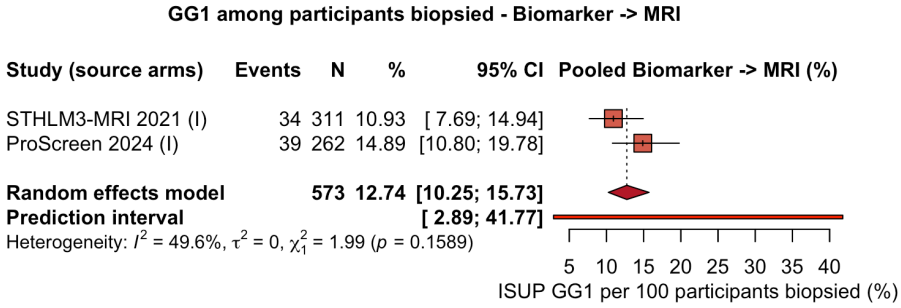


Figure 11c. Forest plot for GG1 among participants biopsied — biomarker → MRI.

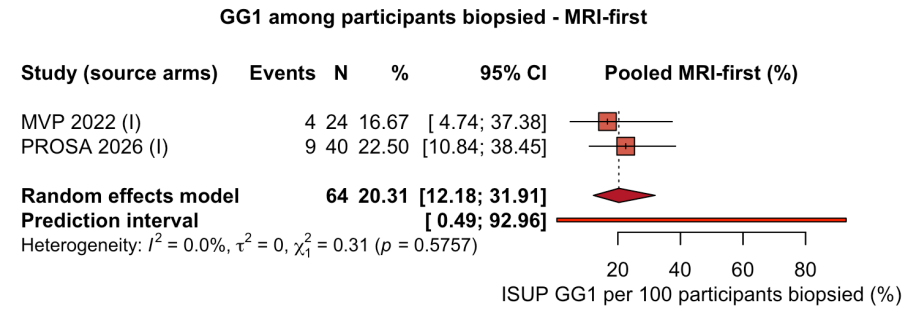


Figure 11d. Forest plot for GG1 among participants biopsied — MRI-first.

4.6 GG≥2 among participants biopsied

Strategy	k	Proportion (95% CI)	I²
PSA	4	17.665% (10.883–27.375)	97.7%
PSA → MRI	2	30.510% (27.526–33.668)	0.0%
Biomarker → MRI	2	61.613% (56.173–66.777)	72.2%
MRI-first	2	43.750% (32.187–56.034)	0.0%

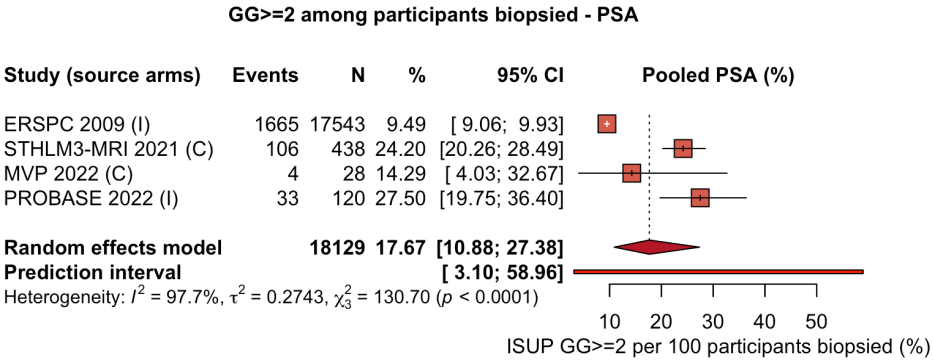


Figure 12a. Forest plot for GG≥2 among participants biopsied — PSA.

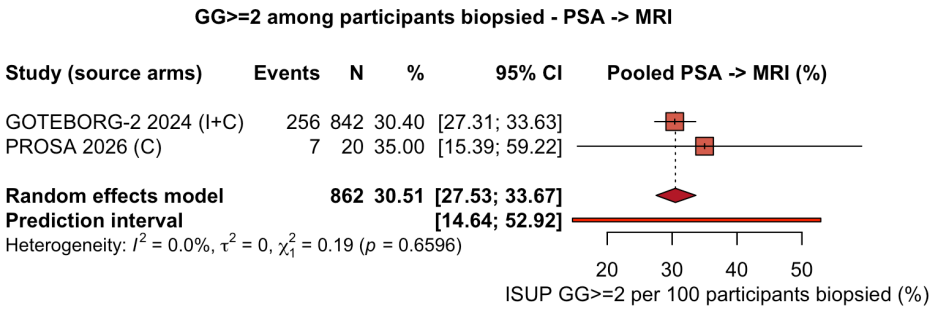


Figure 12b. Forest plot for GG≥2 among participants biopsied — PSA → MRI.

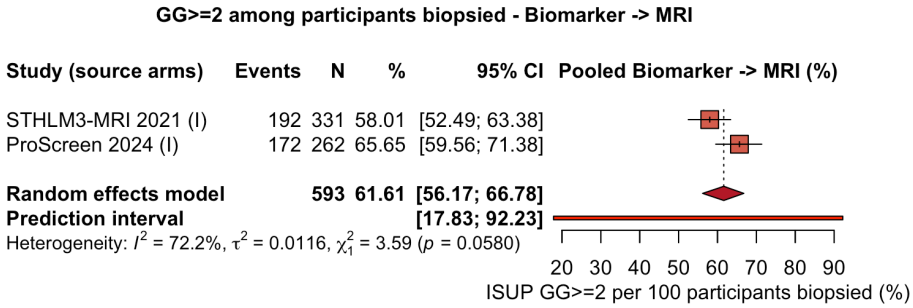


Figure 12c. Forest plot for GG≥2 among participants biopsied — biomarker → MRI.

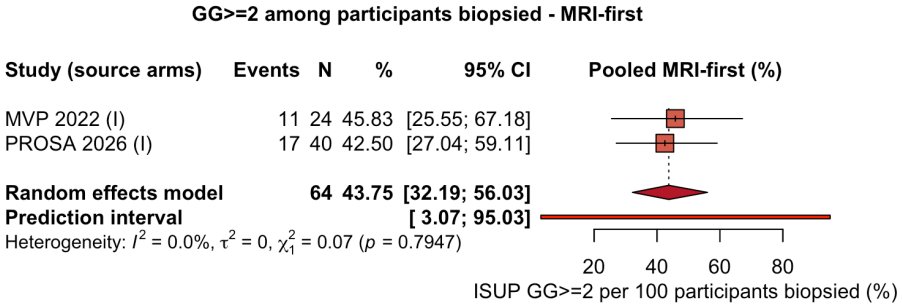


Figure 12d. Forest plot for GG≥2 among participants biopsied — MRI-first.

4.7 Biopsies per screened/assessed participant

Strategy	k	Proportion (95% CI)	I²
PSA	5	6.811% (2.018–20.600)	100.0%
PSA → MRI	2	6.368% (5.969–6.792)	0.0%
Biomarker → MRI	2	3.724% (3.266–4.244)	81.1%
MRI-first	2	10.407% (8.229–13.079)	0.0%

This outcome uses screened/assessed rather than randomized/invited denominators. For ProScreen, the calculation is 262/7,744 (3.38%), not 262/15,201 (1.7%).

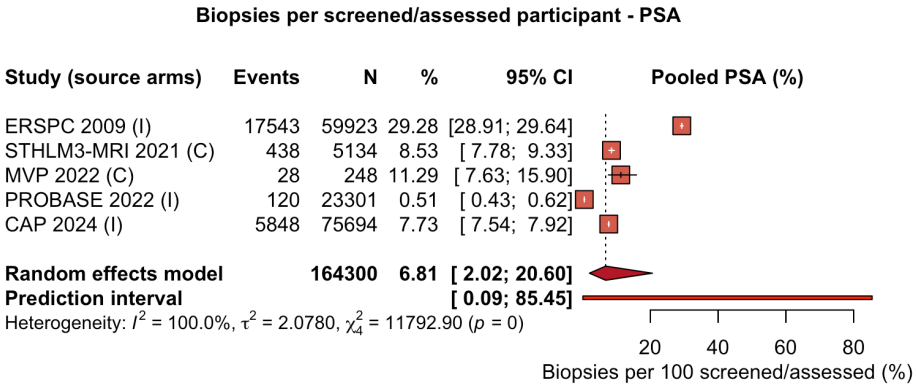


Figure 13a. Forest plot for Biopsies per screened/assessed participant — PSA.

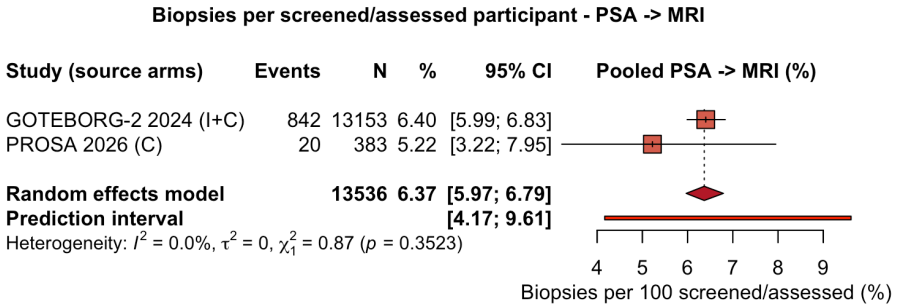


Figure 13b. Forest plot for Biopsies per screened/assessed participant — PSA → MRI.

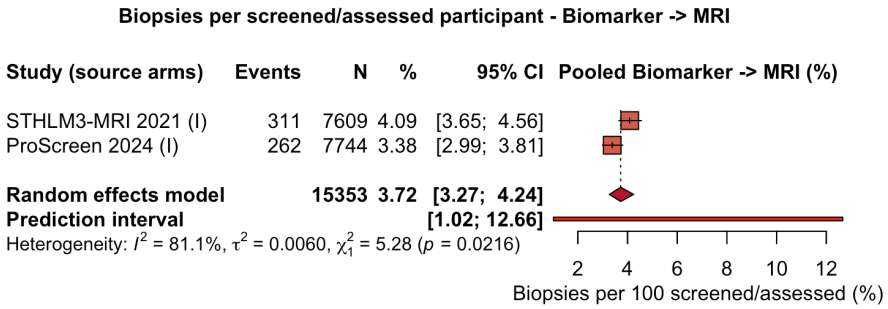


Figure 13c. Forest plot for Biopsies per screened/assessed participant — biomarker → MRI.

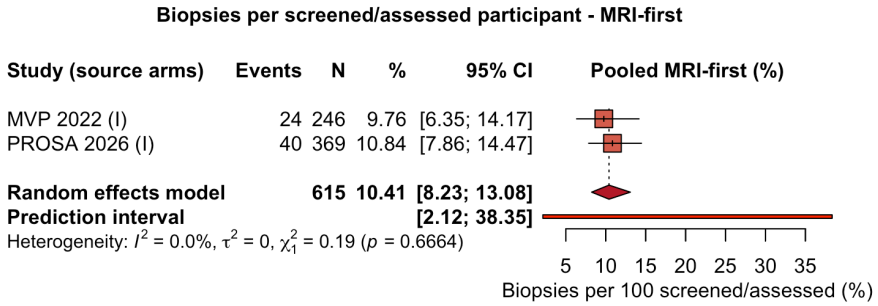


Figure 13d. Forest plot for Biopsies per screened/assessed participant — MRI-first.

5. Sensitivity analyses

Analysis	Outcome	k	Estimate (95% CI)
Invitation cohort excluding CAP	GG1 / randomized	3	RR 1.718 (0.817–3.611)
Invitation cohort excluding CAP	GG1 / all cancers	3	RR 1.162 (0.820–1.648)
Invitation cohort excluding CAP	GG≥2 / randomized	3	RR 1.209 (0.534–2.735)
Invitation cohort excluding CAP	GG≥2 / all cancers	3	RR 0.801 (0.438–1.464)
Invitation cohort excluding ERSPC	GG1 / randomized	3	RR 1.336 (1.071–1.666)
Invitation cohort excluding ERSPC	GG1 / all cancers	3	RR 1.152 (0.862–1.541)
Invitation cohort excluding ERSPC	GG≥2 / randomized	3	RR 1.168 (0.489–2.786)
Invitation cohort excluding ERSPC	GG≥2 / all cancers	3	RR 0.902 (0.759–1.073)
MVP/PROSA assessed denominators	GG1 / assessed	4	RR 0.562 (0.152–2.083)
MVP/PROSA assessed denominators	GG≥2 / assessed	4	RR 1.291 (0.572–2.913)

PROBASE standalone GG≥2 continuity-correction sensitivity:

Added value	RR (95% CI)
0.1	330.433 (0.666–164009.952)
0.5	66.885 (4.098–1091.529)
1.0	33.942 (4.647–247.929)

6. Additional checks and limitations

The analysis generated 131 leave-one-out refits. Complete results are in output/table3_leave_one_out.csv.

Publication-bias analysis was not performed because the largest model contains five studies, below the protocol threshold of ten. Curative treatment, procedural harms, adverse events, RoB 2, and PRISMA flow were not analyzed.

7. Output files

- output/table1_descriptives.csv: arm-level data.
- output/table1b_study_strategy_units.csv: strategy-proportion inputs.
- output/table2_pooled_estimates.csv: pooled estimates, heterogeneity, and prediction intervals.
- output/table3_leave_one_out.csv: leave-one-out results.
- `output/fig1_*` – `output/fig6_*`: comparative and timing RR forest plots.
- `output/fig7_*` – `output/fig13_*`: strategy-proportion forest plots.